

CSC 2201-HUMAN COMPUTER INTERACTION

KK Super Mart Mobile App

Convenience Store / Retail App System | HCI Usability Analysis & Redesign

System Evaluated: KK Super Mart Malaysia App (iOS)**Field Observation:** KK Super Mart De Centrum Branch

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Task 1: User, Task & Human Factors Analysis

1.1 System Overview

The KK Super Mart mobile application (KK Rewards) is the official loyalty and rewards app for KK Super Mart, one of Malaysia's largest 24-hour convenience store chains. Users can register as members, display a digital membership barcode at checkout, collect KK Coins, redeem vouchers, perform daily check-ins for rewards, and view ongoing promotions.

1.2 Field Observation — KK Super Mart De Centrum Branch

A field visit was conducted to observe real user behaviour and gather first-hand evidence for this analysis.

Observation Area	Finding
User Demographics	Mixed ages observed using the app at the cashier, including elderly customers who required staff assistance
Primary In-Store Use	Scanning membership barcode at checkout, checking vouchers, browsing promotions — all three were observed
Barcode Issue	Multiple customers struggled to locate and display the barcode quickly, causing visible delays at the cashier queue
Rewards Awareness	Several customers were unaware of the daily check-in and voucher features buried below promotional banners
Staff Assistance	Staff members were regularly called to assist customers in navigating the app, especially elderly users
Staff Interview	Staff confirmed: customers frequently ask for barcode help; many do not know about rewards; elderly users find navigation confusing

1.3 Target Users

User Group	Age	Tech Literacy	Primary Goal in App
Young Adults / Students	18–25	High	Collect KK Coins, redeem vouchers, and check daily deals
Working Adults	26–45	Medium–High	Loyalty rewards, quick checkout, member exclusive deals
Elderly Customers	55+	Low	Scan membership barcode, redeem discounts at the cashier

1.4 User Personas

Two personas were developed based on field observations and represent the primary and most vulnerable user groups.

Adam Faris University Student Daily KK Mart Visitor Age: 22 Tech Literacy: High	
<i>"I just want to scan and go. Why do I have to tap the screen twice just to show my barcode?"</i>	
Goals • Collect KK Coins on every purchase • Redeem vouchers to save money • Quickly scan the barcode at the cashier without delay	Frustrations • Annoyed when the barcode takes multiple taps to appear • Misses daily check-in because it is hidden under promotions • Finds the home screen cluttered and distracting

Puan Rohani Homemaker Weekly KK Mart Shopper Age: 58 Tech Literacy: Low	
<i>"I always have to ask the cashier to help me find my card in the app. It is embarrassing."</i>	
Goals • Show membership card at checkout to get member price • Occasionally redeem vouchers her daughter tells her about	Frustrations • Cannot find the barcode without staff help • Confused by icons that look similar but do different things • Bright screen is uncomfortable during evening shopping trips

1.5 Task Analysis — User Journey Flow

The diagram below maps the five core user tasks observed in the field, highlighting which steps contain usability issues.

STEP	USER ACTION	USABILITY ISSUE
Step 1	Open app → Register account → Receive OTP verification	Issue 1: No resend button — user stuck if OTP fails
Step 2	Navigate home screen → Find daily check-in & vouchers	Issue 4: Promotions bury rewards features
Step 3	Tap the bottom navigation → Identify the correct tab	Issue 2: Mixed icon styles, no active state
Step 4	Navigate to KK Coins → Tap screen again → Barcode appears	Issue 3: Hidden double-tap, non-discoverable

Step 5	Use the app at night / low light	Issue 5: No dark mode, eye strain
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Figure 1.1 — User Task Flow Diagram showing steps and associated usability issues

1.6 Human Factors Analysis

Factor	Category	Description
Hidden Barcode Gesture	Cognitive — Memory Load	Users must recall a non-obvious double-tap interaction. This violates the 'recognition over recall' principle and increases cognitive load, especially for elderly users.
Mental Model Mismatch	Cognitive — Expectation	Users expect tapping a rewards icon to immediately show their reward. The extra tap required does not match this natural mental model.
Promotions Over Features	Cognitive — Attention	Large promotional banners capture visual attention and direct users away from functional features like check-in and vouchers.
Inconsistent Nav Icons	Perception — Visual Clarity	Mixed solid/stroke icon styles and no highlighted active state reduce the user's ability to quickly identify which section they are in.
No Active Tab Indicator	Perception — Feedback	Without a clear visual response to tapping a navigation item, users receive no confirmation that their action registered.
Elderly Interaction Speed	Interaction — Motor	Elderly users exhibit slower tap speeds and need larger, clearly labelled touch targets. The current hidden gestures are especially problematic for this group.
Night-time Usage	Perception — Environmental	KK Mart is a 24-hour store. A white-only interface causes eye strain in low-light

		environments, reducing usability for a significant portion of users.
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Task 2: Usability Evaluation

Five usability issues were identified through direct app usage, field observation at KK Super Mart De Centrum Branch, and staff interviews. Each issue is evaluated against Nielsen's 10 Usability Heuristics.

2.1 Severity Overview

#	Usability Issue	Severity	Heuristic Violated
#1	WhatsApp OTP Registration Block — users are permanently stuck if the code fails to send	HIGH	<i>Error Prevention</i>
#2	Confusing Navigation — mixed icon styles, no active tab feedback	MEDIUM	<i>Consistency & Standards</i>
#3	Buried Barcode — requires non-discoverable double-tap at cashier	HIGH	<i>Flexibility & Efficiency</i>
#4	Excessive Scrolling — heavy promotions bury rewards and check-in features	HIGH	<i>Aesthetic & Minimalist Design</i>
#5	No Dark Mode — bright interface causes eye strain for 24-hour users	MEDIUM	<i>User Control & Freedom</i>

Figure 2.1 — Usability Issues Severity Overview

2.2 Detailed Usability Issues

#	Issue	HCI Principle Violated	Severity	Justification & Field Evidence
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1	WhatsApp OTP Registration Block: The app only offers WhatsApp verification during sign-up. If the message fails to send, there is no resend button, no countdown timer, and no alternative users cannot complete registration.	Error Prevention	High	Observed during app testing. The original 'Register 2' screen shows no resend option. New users are permanently blocked if WhatsApp delivery fails, preventing access to the entire system. To resolve this critical block, a "Resend OTP" button and an SMS verification alternative have now been added to the design.
2	Confusing Navigation Feedback: The bottom navigation mixes solid and stroke icon styles with varying colours. Tapping an icon gives no visible active state or confirmation of which page is currently open.	Consistency and Standards	Medium	Observed in the app. Staff at De Centrum confirmed customers frequently tap the wrong tabs and lose their place. Inconsistent icons violate the principle of a uniform, predictable interface.
3	The Buried Barcode: To show the membership barcode, a user must go to the KK Coins tab and then tap the screen a second time. This two-step hidden interaction is completely non-	Flexibility and Efficiency of Use	High	Directly observed during the field visit. Staff confirmed they assist customers with this daily. The barcode is the most used feature in-store, yet it is the hardest to access.

	discoverable without prior knowledge.			
4	Excessive Navigation for Rewards & Vouchers: Accessing daily rewards and vouchers requires multiple, non-intuitive steps. To claim a reward, users must first tap "KK Coins" and then scroll down. Accessing vouchers is even more tedious, requiring users to tap "KK Coins," scroll down, tap a separate "Voucher" option, and then finally view them.	Aesthetic and Minimalist Design	High	Staff confirmed that many regular customers do not know about the daily check-in and voucher features. The promotional content competes for attention and wins, hiding functional value from users.
5	No Dark Mode: The app uses a bright white interface throughout with no dark mode option. KK Mart is a 24-hour store, and many customers, especially late night and shift workers, use the app in low-light environments.	User Control and Freedom; Accessibility	Medium	KK Mart's 24-hour operation means a significant portion of users access the app at night. A purely bright interface causes eye strain and reduces the comfort of use for these users.

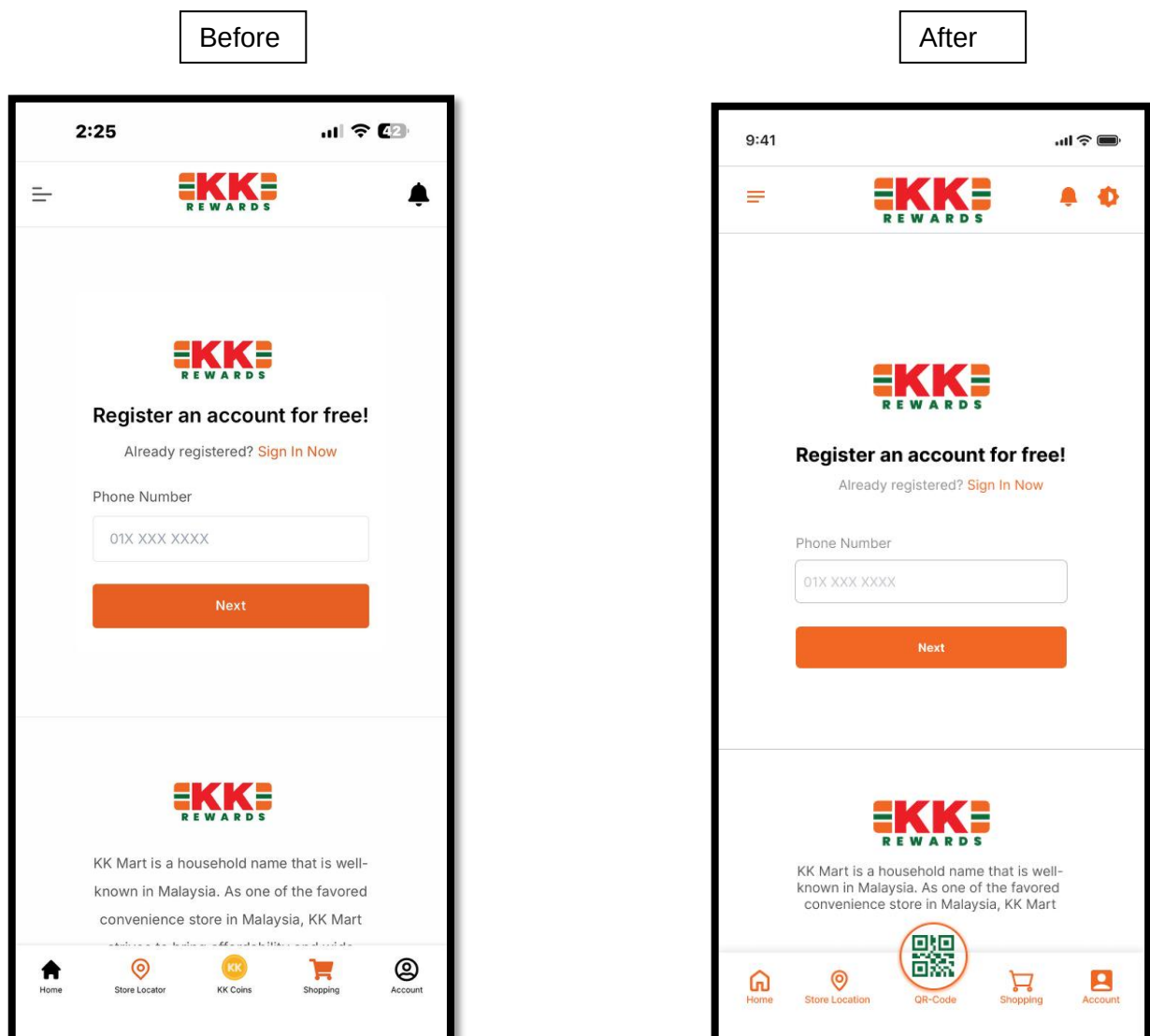
Task 3: Redesign (Prototype)

A high-fidelity prototype was developed using Figma, successfully addressing all five identified usability issues. Both light mode and dark mode versions were designed to accommodate user preferences. The redesign is presented below with before-and-after comparisons.

You can view the full designs and interactive prototype via the links below:

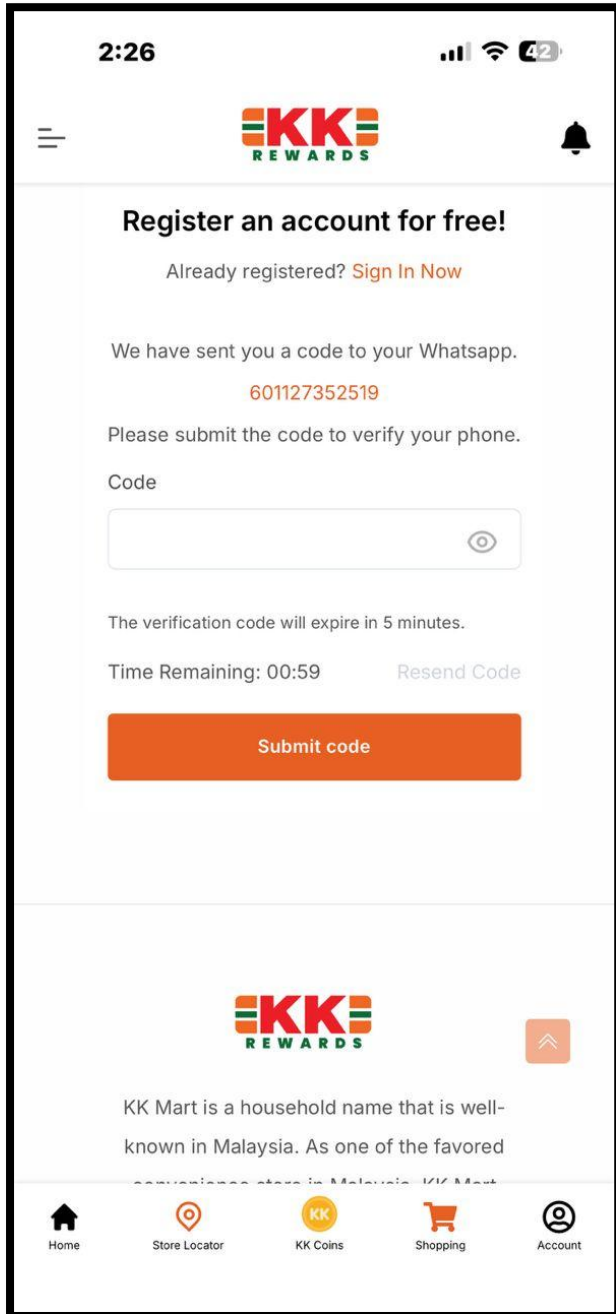
- **Interactive Figma Prototype:** [\[Click Here\]](#)
- **Main Page Design:** [\[Click Here\]](#)

3.1 Before Original KK Mart App Screens and after Redesigning

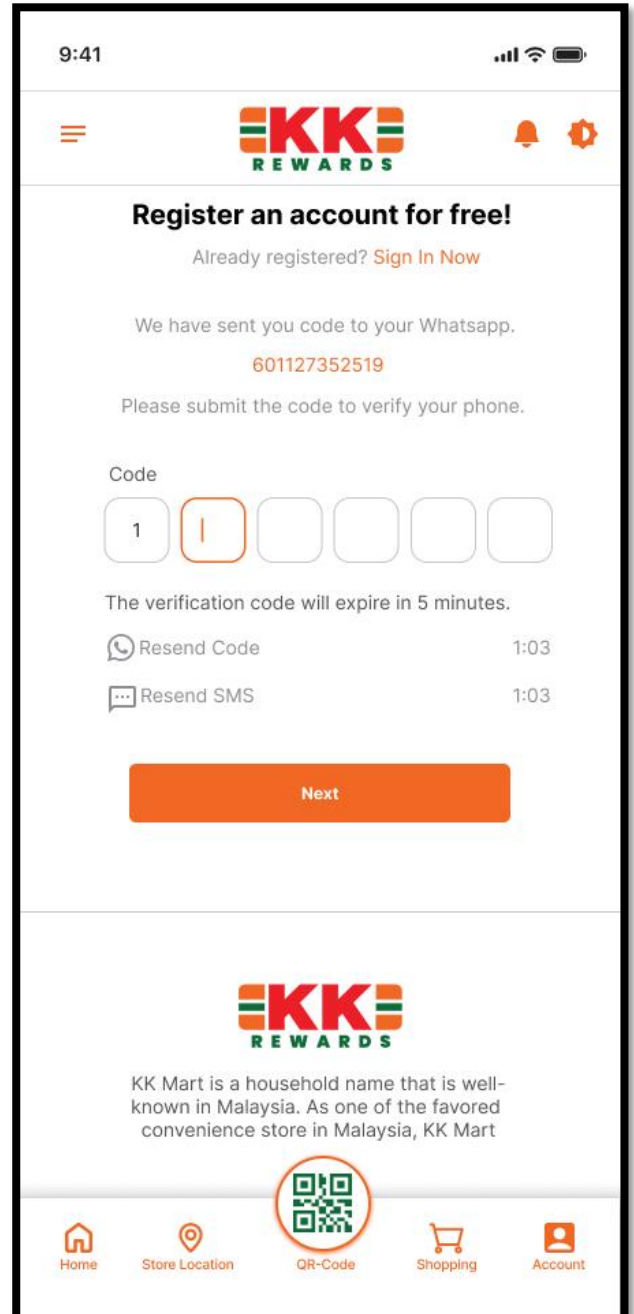


Issue 1

Before

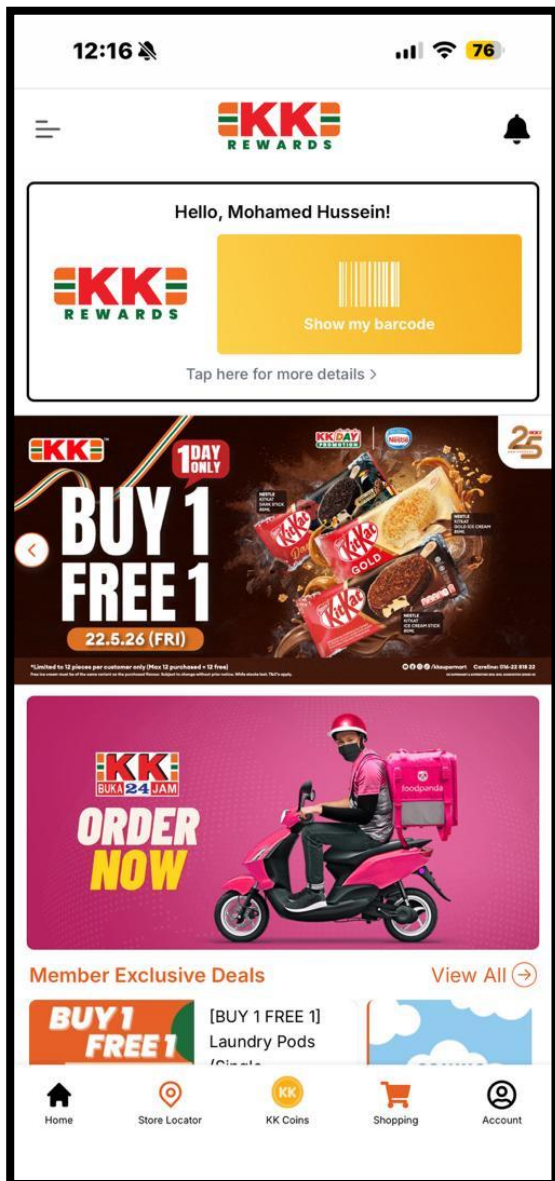


After

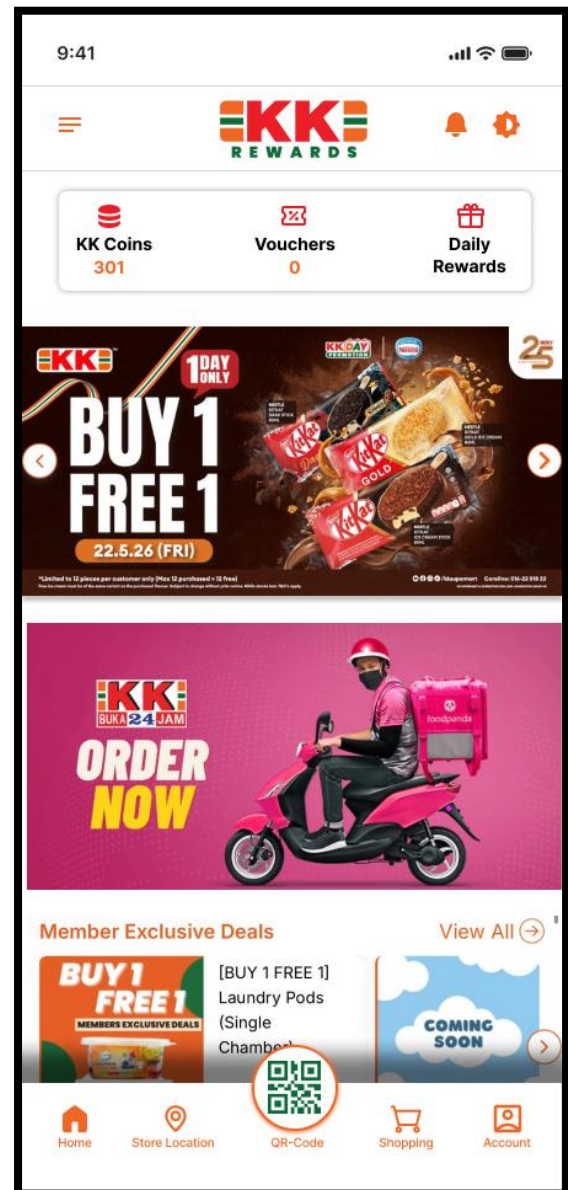


Issue 2

Before



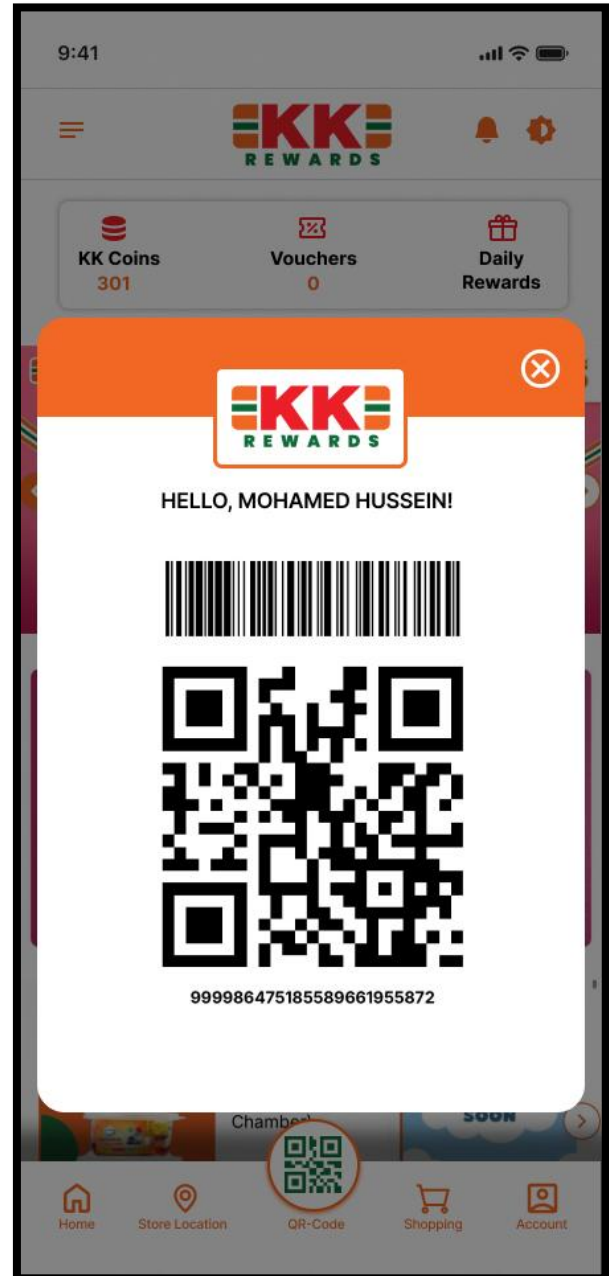
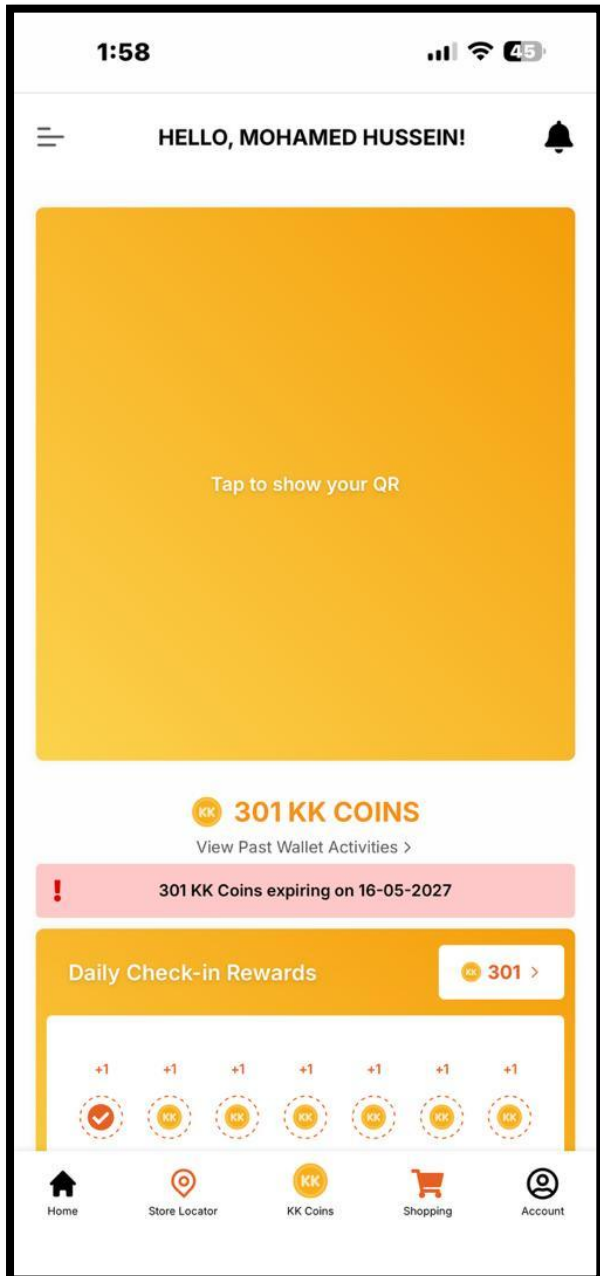
After



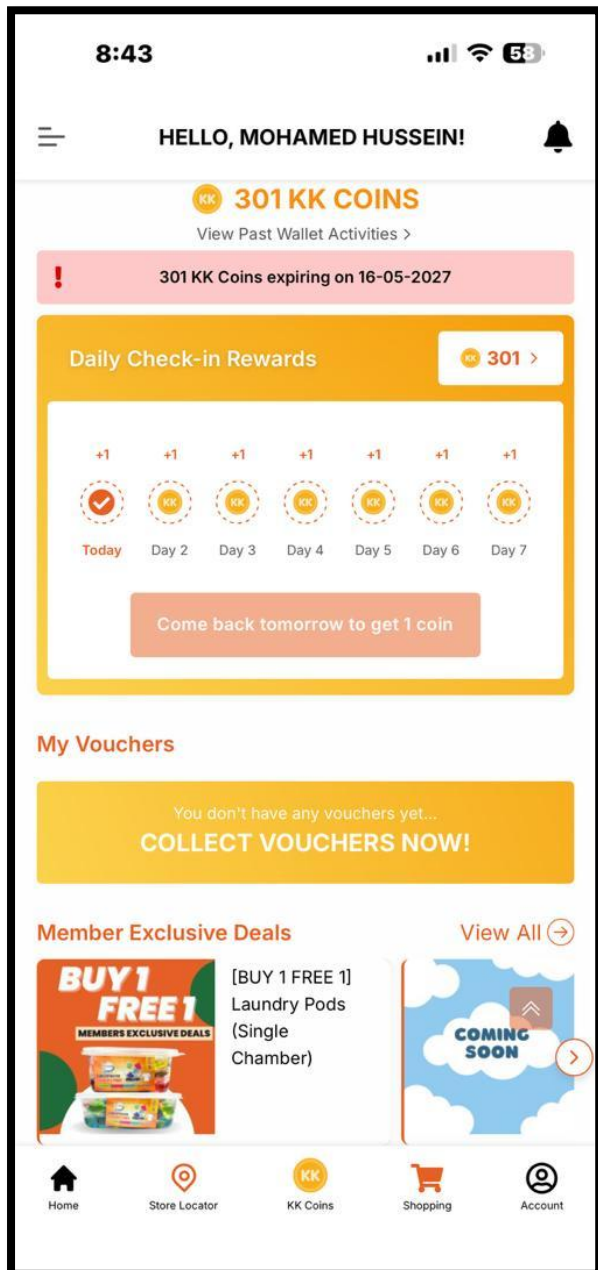
Issue 4

Before

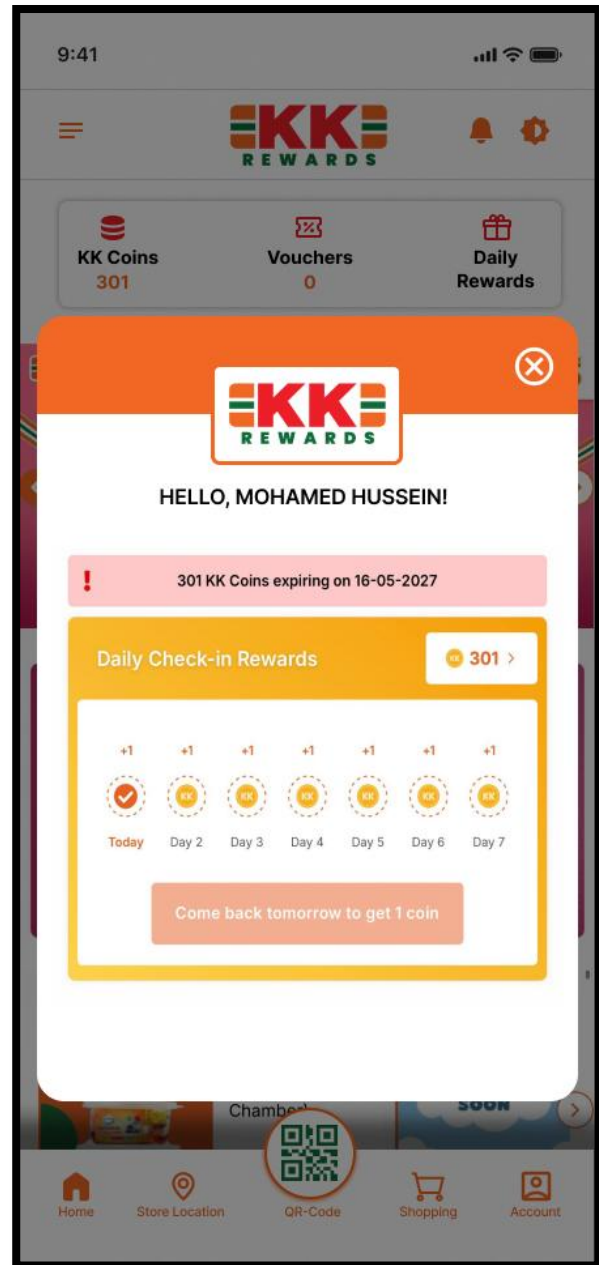
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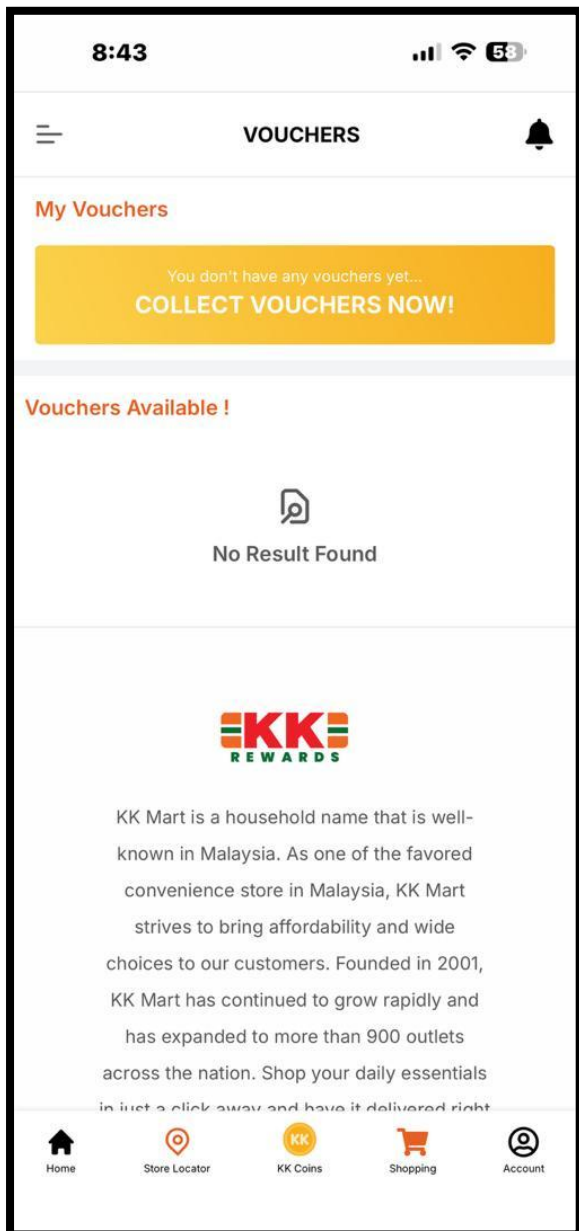
Before



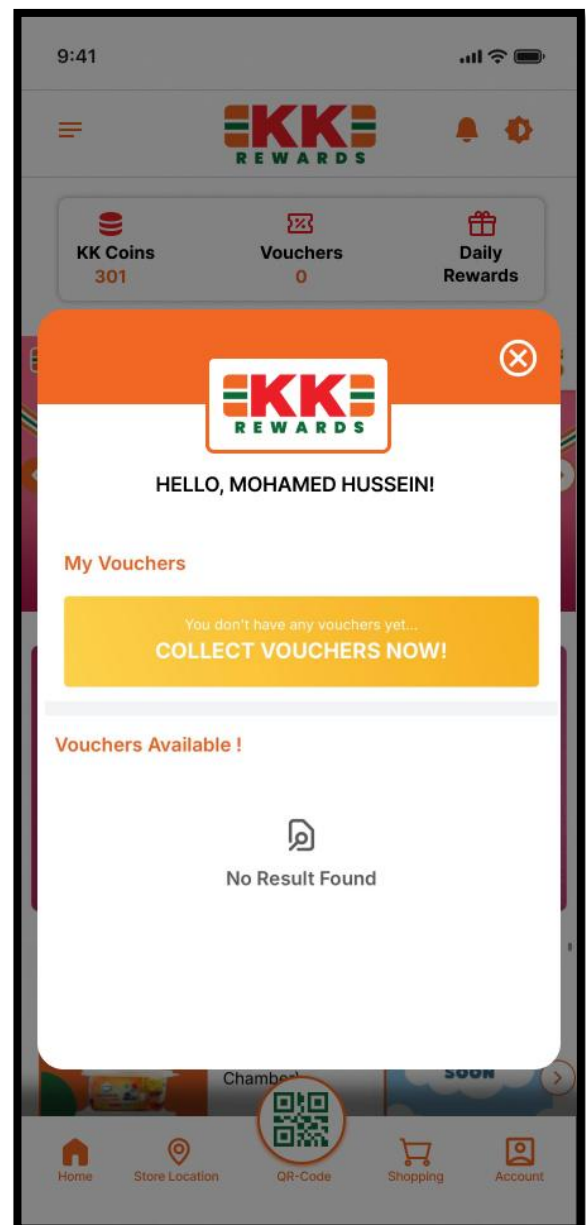
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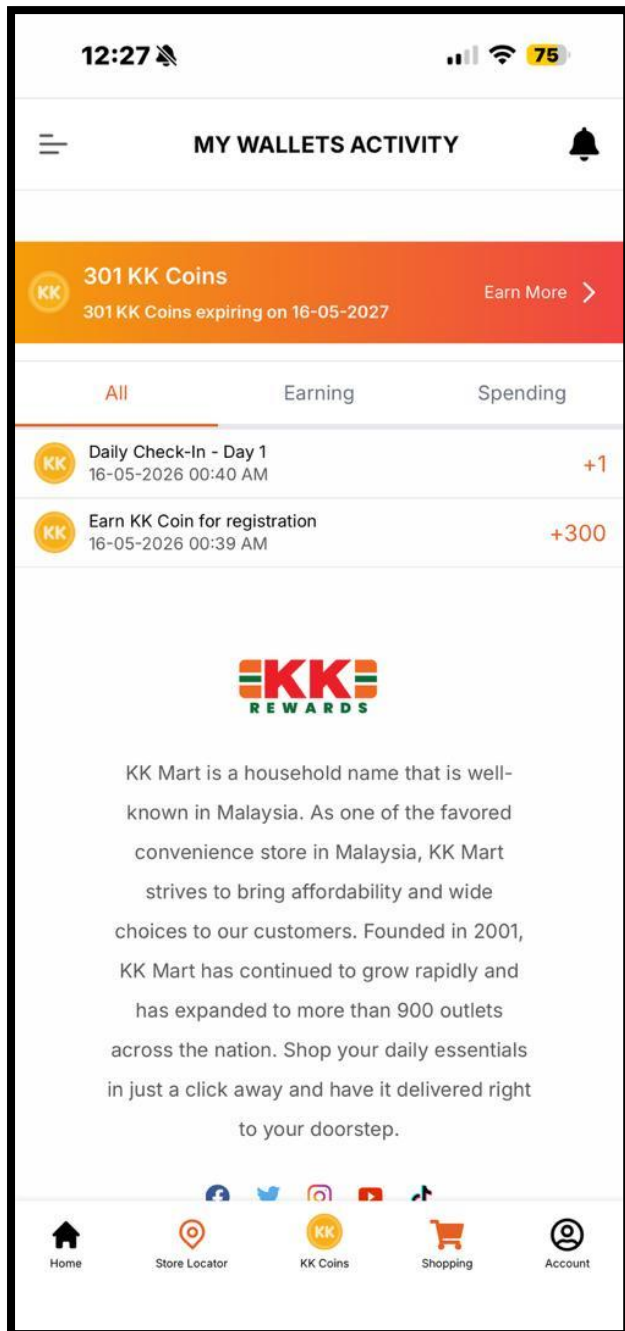
Before



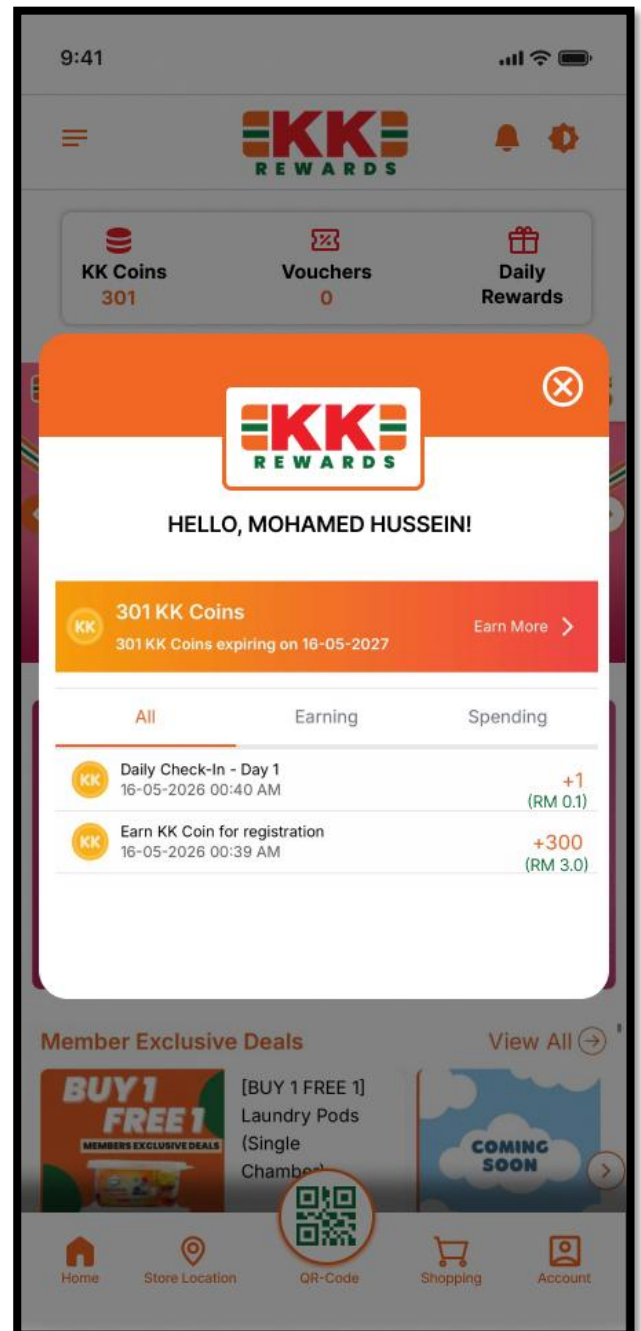
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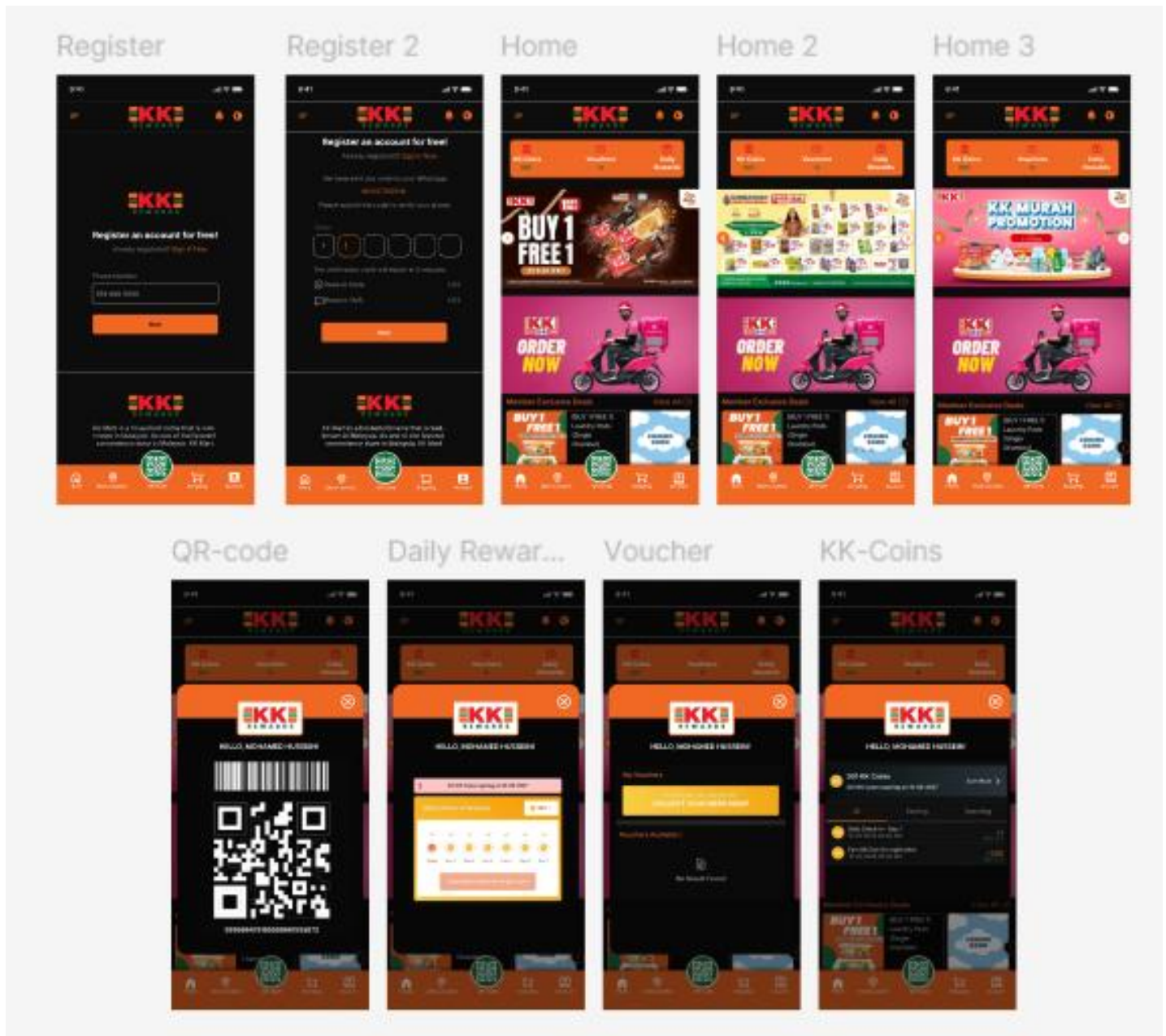
Before



After



Issue 5



3.2 Full Prototype Overview

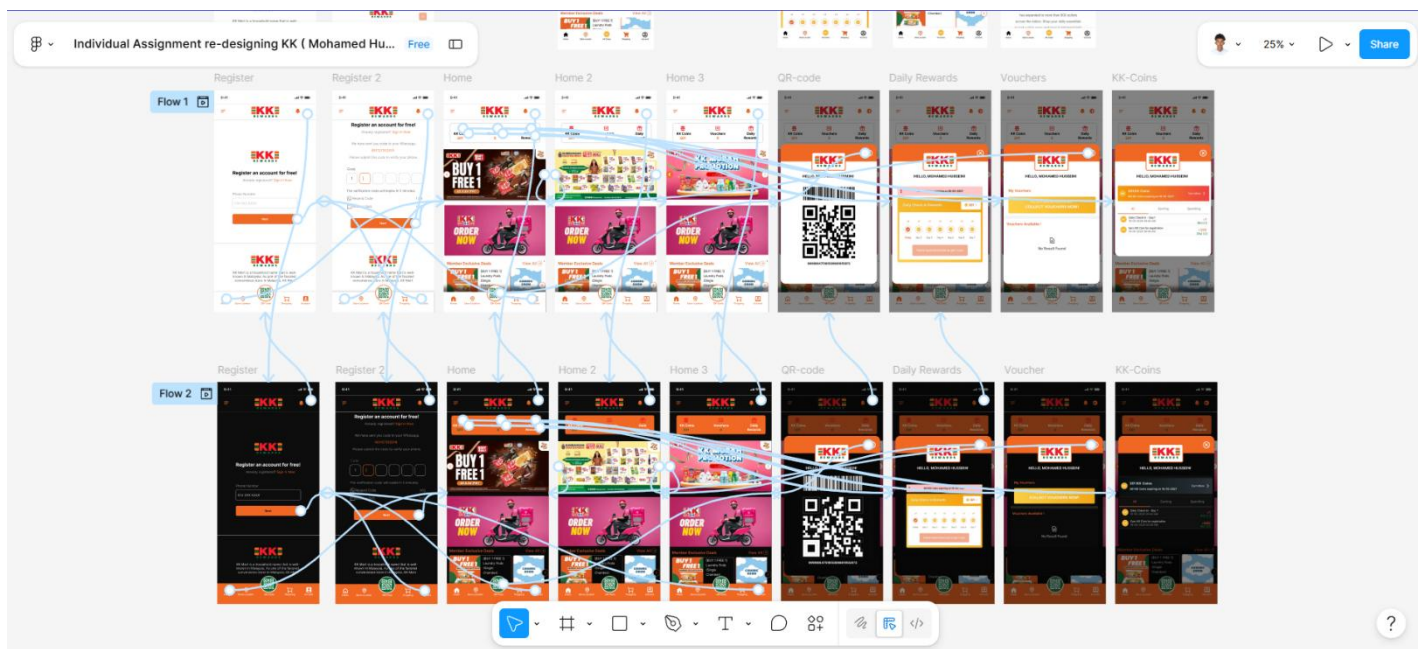
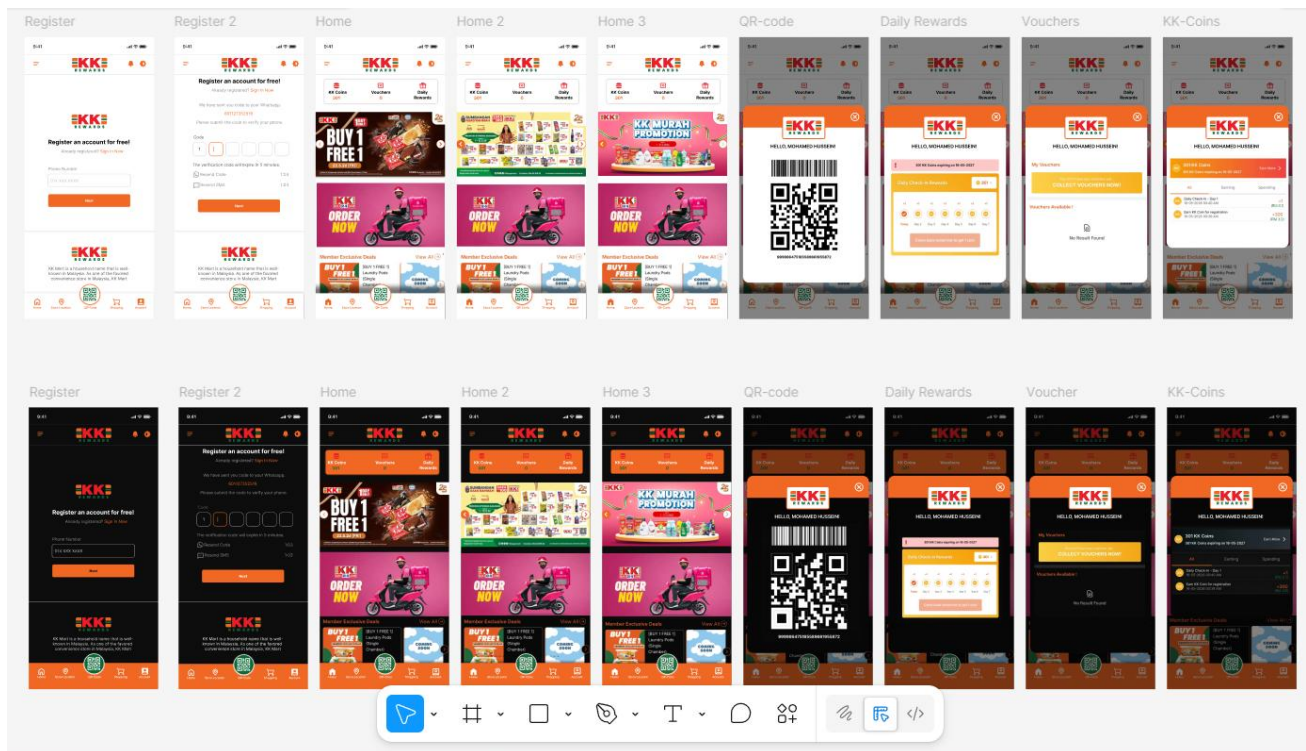


Figure 3.3 — Full Before/After Comparison: Original (top rows) vs Redesigned Light Mode (middle) vs Redesigned Dark Mode (bottom)

3.4 Redesign Decisions

Issue Resolved	Original Problem	Redesign Solution
Issue 1 — OTP Block	WhatsApp only, no SMS, no resend, no timer	Segmented OTP boxes + countdown timer + 'Resend SMS' and WhatsApp options both shown
Issue 2 — Navigation	Mixed icon styles, no active state	Unified solid icons; active tab highlighted in brand red; elevated central QR button added
Issue 3 — Barcode	Hidden double-tap in KK Coins tab	Elevated central QR button on nav bar — one tap from anywhere instantly shows barcode
Issue 4 — Home Hierarchy	Promotions dominate, rewards buried	Action-first layout: coins balance and shortcuts above fold; promotions below
Issue 5 — Dark Mode	Bright white interface only	Full dark mode across all screens; QR box intentionally kept white for scanner compatibility

3.5 Key Design Insight — Hardware Aware Dark Mode

The most technically thoughtful decision in this redesign is the treatment of the barcode screen in dark mode. While all other screens use a dark background, the membership barcode is intentionally placed inside a pure white rectangle even when dark mode is active.

This is because physical barcode scanners at KK Super Mart cashier counters require high contrast between the barcode and its background to function correctly. A dark background would cause scanner failure. By keeping only the barcode container white in all themes, the redesign ensures that the accessibility improvement (dark mode) never conflicts with the core function of the app.

3.6 Screen-by-Screen Prototype Description

Screen	Issue Solved	What Changed
Register	Issue 1	Segmented OTP input boxes (one per digit), countdown timer, Resend SMS, and WhatsApp links clearly shown below the code entry field
Home	Issue 4	KK Coins balance and wallet info elevated to the top, a quick-access row for Vouchers and Daily Check-in added, promotions pushed below
Bottom Navigation	Issues 2 & 3	Icons unified to a consistent solid style, active tab shown in red, elevated central QR button provides one-tap barcode access from any screen
QR Code Screen	Issue 3	Barcode appears instantly on tap, displayed in a pure white box in both light and dark mode to ensure store scanner compatibility
Daily Rewards	Issue 4	Directly accessible from home, quick-access row, no longer buried under promotional content
Vouchers	Issue 4	Directly accessible, Collect Vouchers Now CTA added to empty state, so users understand how to earn vouchers
KK Coins	Issue 4	Coins balance and RM equivalent are shown clearly, transaction history is visible at a glance
Dark Mode (all screens)	Issue 5	Deep dark background applied across all screens; brand red used as accent, QR box remains white for hardware compatibility

Task 4: Final Evaluation & Justification

This section compares the original and redesigned KK Super Mart app and justifies how each change addresses the identified usability issues using HCI principles.

4.1 Before vs After Comparison

Issue	Before	After	HCI Principle Applied
OTP Registration Block	WhatsApp only, no resend, users are permanently stuck	Segmented OTP + timer + Resend SMS + WhatsApp options	Error Prevention
Navigation Confusion	Mixed icons, no active state, wrong tabs tapped	Unified icons, red active state, elevated QR button	Consistency and Standards
Buried Barcode	Double-tap hidden interaction, cashier delays; staff help needed	One-tap access via central nav button, instant display	Flexibility and Efficiency of Use
Excessive Scrolling	Promotions bury rewards, users are unaware of check-in/vouchers	Action-first home, coins and shortcuts visible immediately	Aesthetic and Minimalist Design
No Dark Mode	White-only, eye strain for 24-hour/night users	Full dark mode, white QR box preserves scanner function	User Control and Freedom

4.2 HCI Principles Applied

- Error Prevention — OTP resend with SMS and WhatsApp options eliminates the registration dead-end. Users can always recover from a failed code.

- **Consistency and Standards** — Unified icon style and colour across all navigation elements. Users build a reliable mental model of the interface.
- **Flexibility and Efficiency of Use** — The elevated central QR button is the single most impactful change. Every user, regardless of experience, can access their barcode in exactly one tap.
- **Aesthetic and Minimalist Design** — Removing promotional dominance from the home screen reduces cognitive overload and surfaces the features users actually need.
- **User Control and Freedom** — Dark mode gives users a choice over their visual environment, especially important for KK Mart's 24-hour customer base.

4.3 Impact on Field-Observed Problems

Field Observation	Root Cause	Resolved By
Staff regularly help customers find the barcode	Hidden double-tap interaction in the KK Coins tab	One-tap elevated QR button on the bottom navigation
Customers are unaware of rewards and vouchers	Promotions dominate the home screen, and features are buried	Action-first home layout with quick-access shortcuts
Elderly users are confused by navigation	Mixed icon styles with no active state feedback	Unified icons with a clear red active state indicator
New users get stuck during registration	WhatsApp-only OTP with no resend option	Segmented OTP with timer, resend SMS, and WhatsApp options
Discomfort using the app at night	No dark mode available	Full dark mode with hardware-aware white QR box

4.4 Conclusion

The redesigned KK Super Mart app successfully resolves all five usability issues identified through real-world field observation, staff interviews, and direct app analysis at KK Super

Mart De Centrum Branch. Every design decision is grounded in HCI principles and validated by real evidence.

The most impactful single change is the elevated central QR button, which transforms a multi-step hidden interaction into a single tap, directly eliminating the cashier queue delays observed in the field. The hardware-aware dark mode implementation demonstrates that strong HCI design considers not just the screen, but the real-world physical context in which the app is used.

Together, these improvements make the KK Super Mart app significantly more accessible, efficient, and usable for all customer groups, from tech-savvy young adults to elderly shoppers who previously required staff assistance for basic tasks.

